

Article

Prescribed Performance Global Non-Singular Fast Terminal Sliding Mode Control of PMSM Based on Linear Extended State Observer

Yifei Yao ^{1,†} , Yaoming Zhuang ^{2,*,†} , Yizhi Xie ² , Peng Xu ² and Chengdong Wu ² 

¹ College of Information Science and Engineering, Northeastern University, Shenyang 110819, China; 2300935@stu.neu.edu.cn

² Faculty of Robot Science and Engineering, Northeastern University, Shenyang 110819, China; 2302181@stu.neu.edu.cn (Y.X.); 2302182@stu.neu.edu.cn (P.X.); wuchengdong@mail.neu.edu.cn (C.W.)

* Correspondence: zhuangyaoming@mail.neu.edu.cn

† These authors contributed equally to this work.

Abstract: In manufacturing, the position tracking accuracy and stability of Permanent Magnet Synchronous Motors are often challenged by uncertainties, especially in complex environments. Existing control methods struggle to balance fast response with high-precision tracking. To address this, we propose a Prescribed Performance Global Non-Singular Fast Terminal Sliding Mode Control (PPGNFTSMC) method using a linear extended state observer (LESO). A smooth and bounded prescribed performance function is designed to ensure finite-time convergence while satisfying performance requirements such as overshoot and settling time. Based on this function, the system error is reconstructed to align the system response with predefined specifications. The reconstructed error is then used to design a global non-singular fast terminal sliding mode surface. A LESO is employed for real-time disturbance estimation, and the disturbance estimates, along with the sliding mode surface, are used to derive the control law for the position–speed integrated controller. Experimental results show that the proposed method outperforms the comparison methods in transient response, tracking accuracy, and robustness across various signal types.

Keywords: position servo control; permanent magnet synchronous motor (PMSM); global non-singular fast terminal sliding mode control; prescribed performance control; linear extended state observer



Academic Editor: Ioan Ursu

Received: 24 December 2024

Revised: 23 January 2025

Accepted: 27 January 2025

Published: 30 January 2025

Citation: Yao, Y.; Zhuang, Y.; Xie, Y.; Xu, P.; Wu, C. Prescribed Performance Global Non-Singular Fast Terminal Sliding Mode Control of PMSM Based on Linear Extended State Observer. *Actuators* **2025**, *14*, 65. <https://doi.org/10.3390/act14020065>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

In modern manufacturing, high-end Computer Numerical Control (CNC) machine tools are vital, directly affecting product quality and production efficiency due to their precision and performance [1]. These machines are commonly employed for precise machining tasks such as high-speed cutting and intricate engraving, which demand exceptional motion control accuracy. For high-end applications, CNC machines must precisely follow complex machining paths, necessitating a high-performance position control system. These demands drive continuous development and innovation in CNC machine tool technology to meet the needs of modern manufacturing. Permanent magnet synchronous motors (PMSMs) are ideal for driving high-end CNC machine tools, given their compact size, fast response, and high efficiency [2]. Nevertheless, the complexity and variability of actual production processes can affect motor behavior due to internal parameter changes and external load disturbances, making it challenging to ensure position-tracking accuracy and

system stability. Therefore, achieving high-position command response tracking control for a PMSM is critical to address the challenges faced by high-end CNC machines.

Currently, various advanced control methods are applied in this field, primarily categorized as follows: active disturbance rejection control [3,4], backstepping control [5], model predictive control [6], fuzzy control [7,8], and sliding mode control (SMC) [9–12]. Among these, SMC is extensively utilized in PMSM position tracking control due to its simplicity, fast response, and strong robustness. However, traditional SMC methods using linear sliding surfaces cannot guarantee that tracking errors converge within a finite time. To address this limitation, various improved SMC strategies have been proposed.

Initially, terminal sliding mode control [13] introduced nonlinear functions into the sliding surface, ensuring that tracking errors converge within a finite time. However, this method suffers from singularity problems and slow convergence. To address these drawbacks, non-singular terminal sliding mode control [14,15] and fast terminal sliding mode control [16] were developed, focusing on optimizing singularity handling and convergence speed, respectively. Nevertheless, these approaches only partially resolved the challenges. Consequently, non-singular fast terminal sliding mode control (NFTSMC) [17–19] emerged, comprehensively addressing these challenges. This strategy ensures finite-time convergence of tracking errors and avoids singularities while enhancing the system's rapidity during the reaching phase. However, nonlinear sliding modes generally converge slower near equilibrium compared to linear ones. Moreover, achieving robustness requires a large switching gain, potentially leading to severe chattering issues. To manage these challenges, observers are commonly used to estimate disturbances for feed-forward compensation, or adaptive laws are integrated. These techniques have been successfully applied to various nonlinear systems such as servo systems and robotic arms. For instance, Liu et al. [20] introduces a composite speed controller that integrates the modified super-twisting sliding-mode controller with an ESO, significantly improving the disturbance rejection capability in surface-mounted PMSM drives. Similarly, Dang et al. [21] presents an adaptive SMC strategy combined with an ESO to enhance the speed control performance of PMSM drives, ensuring faster dynamic responses and better robustness against internal and external disturbances. While these approaches effectively address some of the limitations of traditional SMC, such as slow convergence and vulnerability to uncertainties, challenges remain, particularly with NFTSMC. Specifically, the robustness of NFTSMC is mainly effective during the sliding phase, which may limit its performance in more complex or highly dynamic systems.

In servo control systems, transient response characteristics such as maximum overshoot and convergence speed are as critical as steady-state error. Recently, a method called prescribed performance control (PPC) [22] was proposed. The core idea is to transform the system error using a prescribed performance function, enabling precise control over system performance by managing the reconstructed error. This method allows the system state to respond quickly and accurately according to predefined performance requirements and has found successful applications across diverse fields [23,24].

Inspired by previous research, the paper proposes a Linear Extended State Observer (LESO)-based Prescribed Performance Global Non-Singular Fast Terminal Sliding Mode Control (PPGNFTSMC) method for PMSM. This paper makes the following key contributions:

1. **Enhanced Precision:** The PPC method is utilized to accelerate the transient response and improve the precision of the PMSM position servo system.
2. **Real-time Disturbance Observation:** A LESO is used for real-time disturbance estimation, and the estimated disturbance information is employed by the controller to provide feed-forward compensation to the control input, ensuring strong anti-disturbance capability with a smaller switching gain.

3. **Dual-loop Structure:** A dual-loop structure is designed for the position servo system controller. This includes an outer position–speed integrated loop with LESO-based PPGNFTSMC and an inner current loop with a PI controller, thereby enhancing tracking accuracy and adaptability.
4. **Integrated Control Design:** A position–speed integrated controller is designed based on the proposed global non-singular fast terminal sliding mode surface (GNFTSMS). The controller incorporates reconstructed error from PPC and disturbance estimates, ensuring a fast global response and strong robustness.

These contributions demonstrate the method’s ability to accelerate transient response, reduce steady-state error, and maintain robust performance under disturbances.

The structure of this paper is as follows: Section 2 introduces the mathematical model of the PMSM. Section 3 explains the design process of the controller. Section 4 presents the experimental results. Finally, Section 5 highlights the contributions and outlines directions for future work.

2. Model Description

Figure 1 illustrates the structure of the PMSM, assuming an ideal surface-mounted PMSM with the following characteristics: (1) non-saturated magnetic circuit; (2) negligible hysteresis and eddy current losses; and (3) spatially sinusoidal distribution of the three-phase stator windings.

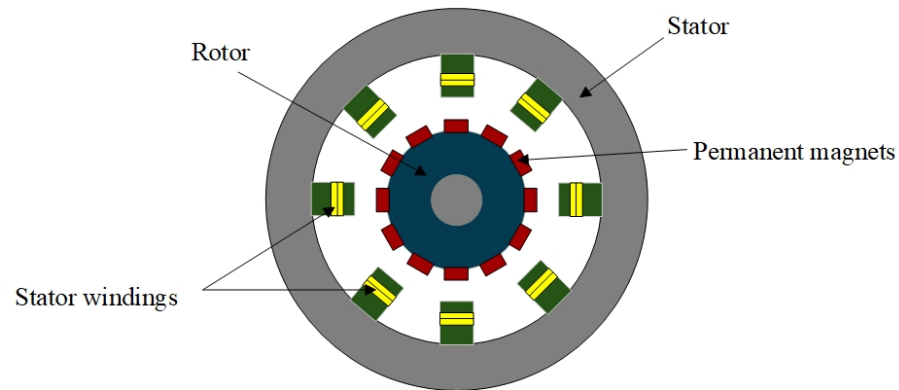


Figure 1. Structural diagram of PMSM.

The dynamic model in the d-q coordinate system is given by:

$$\begin{cases} u_d = L_d \frac{di_d}{dt} + Ri_d - p\omega L_q i_q \\ u_q = L_q \frac{di_q}{dt} + Ri_q + p\omega (L_d i_d + \Psi_f) \end{cases} \quad (1)$$

where u_d and u_q denote the stator voltage of the d-q axes, L_d and L_q denote the stator inductances of the d-q axes, i_d and i_q denote the stator current of the d-q axes, R is the per-phase stator resistance, p is the number of pole pairs, ω is the mechanical angular speed of the rotor, and Ψ_f is the rotor flux linkage.

The dynamic model is described as:

$$\dot{\theta} = \omega \quad (2)$$

$$J \frac{d\omega}{dt} = T_e - T_L - B\omega \quad (3)$$

where θ is the rotor position, J is the moment of inertia, T_e is the electromagnetic torque, T_L is the load torque, and B is the viscous friction coefficient.

The equation for electromagnetic torque is given as:

$$T_e = \frac{3}{2} p i_q [i_d (L_d - L_q) + \Psi_f] \quad (4)$$

To maintain constant flux in a PMSM, the control strategy often sets $i_d = 0$. For a surface-mounted PMSM, $L_d = L_q$, leading to:

$$T_e = \frac{3}{2} p \Psi_f i_q \quad (5)$$

However, during actual operation, the servo control system is influenced by uncertainties, including internal parameter variations, external load disturbances, and unmodeled dynamics. Assuming $J = J_n + \Delta J$, where J_n is the rated value of J , the dynamic model can be rewritten as:

$$\ddot{\theta} = \dot{w} = b i_q + d \quad (6)$$

where $b = \frac{3p\Psi_f}{2J_n}$, and $d = -\frac{1}{J_n}(B\dot{w} + T_L + \Delta J \frac{dw}{dt})$ is a nonlinear function that characterizes the dynamics and external disturbances of the system, which is collectively referred to as 'disturbance'. In this study, it is assumed that d is a slowly varying quantity, as external load disturbances T_L , parameter variations ΔJ , and friction torque $B\dot{w}$ typically change gradually over time in practical systems. Based on this assumption, $\dot{d}$ can be approximated as zero.

3. Controller Design

Figure 2 illustrates the overall system structure. The specific design process is as follows:

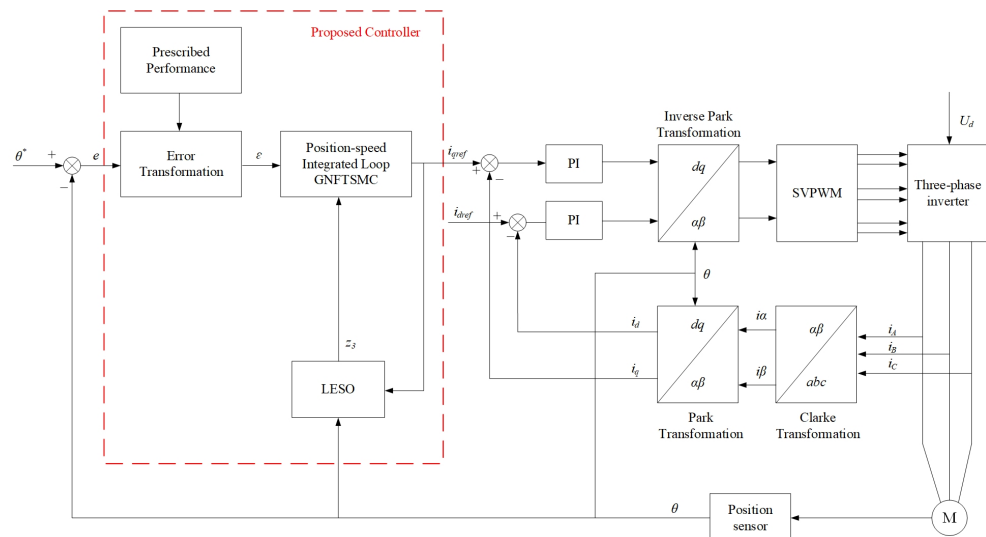


Figure 2. Overall block diagram of the system.

3.1. Prescribed Performance Function

To enhance PMSM position servo system performance, a smooth prescribed performance function (PPF) $\mu(t) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is introduced to define and constrain the system performance indices, ensuring desired dynamic and static performance. The PPF should satisfy as follows:

- (1) $\mu(t)$ is positive and decreasing.
- (2) $\lim_{t \rightarrow \infty} \mu(t) = \mu_\infty > 0$.

The PPF is chosen to be:

$$\mu(t) = (\mu_0 - \mu_\infty)e^{-lt} + \mu_\infty \quad (7)$$

where μ_0 is the initial value, μ_∞ is the upper limit of the steady-state error, and l is the convergence rate, all positive values with $\mu_0 > \mu_\infty$.

The tracking error $e(t)$ satisfies:

$$-\underline{\delta}\mu(t) < e(t) < \bar{\delta}\mu(t) \quad \forall t > 0 \quad (8)$$

where $\underline{\delta}$ and $\bar{\delta}$ are positive constants, chosen by the designer, and are associated with the maximum overshoot.

To prevent the original nonlinear system's tracking error from being affected by the constraints defined in the above Equation (8), an error transformation method is introduced, converting it into an equivalent unconstrained system. Define:

$$e(t) = \mu(t)S(\varepsilon) \quad (9)$$

where ε is the reconstructed error, and $S(\varepsilon)$ is a smooth and strictly increasing function with properties as follows:

- (1) $-\underline{\delta} < S(\varepsilon) < \bar{\delta}, \forall \varepsilon \in L_\infty$.
- (2) $\lim_{\varepsilon \rightarrow +\infty} S(\varepsilon) = \bar{\delta}$, and $\lim_{\varepsilon \rightarrow -\infty} S(\varepsilon) = -\underline{\delta}$.

Choose $S(\varepsilon)$ such that it satisfies the above characteristics as follows:

$$S(\varepsilon) = \frac{\bar{\delta}e^\varepsilon - \underline{\delta}e^{-\varepsilon}}{e^\varepsilon + e^{-\varepsilon}} \quad (10)$$

Furthermore, applying the inverse transformation to Equation (10) based on the properties of $S(\varepsilon)$ yields the reconstructed error as follows:

$$\varepsilon = S^{-1}\left[\frac{e(t)}{\mu(t)}\right] = \frac{1}{2} \ln \frac{z(t) + \bar{\delta}}{\bar{\delta} - z(t)} \quad (11)$$

where $z(t) = \frac{e(t)}{\mu(t)}$.

Thus, ensuring the bounded reconstructed error ε can guarantee Equation (8).

3.2. Linear Extended State Observer Design

In practical applications, disturbances are often unknown. The LESO plays a crucial role in estimating unmeasurable states in real time. Compared to traditional Extended State Observers (ESOs), the LESO simplifies the observer design through linearization while maintaining robustness to model uncertainties and external disturbances. The primary advantages of the LESO include reduced computational complexity, improved real-time performance, and easier parameter tuning, making it particularly well suited for high-precision servo systems and practical engineering applications. In this work, the proposed LESO addresses the challenges of disturbance estimation and system state reconstruction, ensuring accurate control and robust performance. Using measurable rotor position information $y = \theta$, the LESO is defined as:

$$\begin{cases} e_\theta = z_1 - y \\ \dot{z}_1 = z_2 - \beta_1 e_\theta \\ \dot{z}_2 = z_3 - \beta_2 e_\theta + bu \\ \dot{z}_3 = -\beta_3 e_\theta \end{cases} \quad (12)$$

where e_θ represents the difference between estimated and actual rotor positions, z_1 , z_2 , and z_3 correspond to the estimated rotor position, estimated rotor mechanical angular velocity, and estimated disturbance, respectively. β_1 , β_2 , and β_3 are the error gains at various orders of the LESO. u is the control input.

The state estimation error is expressed as follows:

$$e_\theta = z_1 - y, \quad e_w = z_2 - \dot{y}, \quad e_d = z_3 - d \quad (13)$$

where e_θ is the position observation error, e_w is the mechanical angular velocity observation error, and e_d is the disturbance observation error.

Based on Equation (12), the state-space description of the observation error is derived as follows:

$$\dot{\mathbf{e}} = \begin{bmatrix} \dot{e}_\theta \\ \dot{e}_w \\ \dot{e}_d \end{bmatrix} = \begin{bmatrix} -\beta_1 & 1 & 0 \\ -\beta_2 & 0 & 1 \\ -\beta_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_\theta \\ e_w \\ e_d \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\dot{d} \end{bmatrix} \quad (14)$$

Let $A = \begin{bmatrix} -\beta_1 & 1 & 0 \\ -\beta_2 & 0 & 1 \\ -\beta_3 & 0 & 0 \end{bmatrix}$, $\mathbf{e} = \begin{bmatrix} e_\theta \\ e_w \\ e_d \end{bmatrix}$, $\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -\dot{d} \end{bmatrix}$, then the dynamic equation of the observation error can be written in matrix form:

$$\dot{\mathbf{e}} = A\mathbf{e} + \mathbf{g} \quad (15)$$

Considering that $\dot{d} \approx 0$, Equation (15) simplifies to the homogeneous system:

$$\dot{\mathbf{e}} = A\mathbf{e} \quad (16)$$

The characteristic equation of this system is derived as follows:

$$f(\lambda) = \det(\lambda I - A) = \lambda^3 + \beta_1\lambda^2 + \beta_2\lambda + \beta_3 = 0 \quad (17)$$

Therefore, to ensure that the observer error converges, the eigenvalues should all lie in the left half of the complex plane. This assumes the characteristic equation of an ideal observer is:

$$f^*(\lambda) = (\lambda + w_0)^3 \quad (18)$$

where w_0 represents the observer bandwidth.

Comparing Equations (17) and (18), it can be deduced that $\beta_1 = 3w_0$, $\beta_2 = 3w_0^2$, $\beta_3 = w_0^3$. By selecting parameters based on the bandwidth criteria, optimizing system performance effectively requires adjusting only one parameter, which simplifies the design and debugging process of the control system.

According to the system's linear stability theory, the observation error $\mathbf{e}$ will exponentially converge to zero:

$$\mathbf{e}(t) \rightarrow 0 \quad t \rightarrow \infty \quad (19)$$

Thus, $z_1 \rightarrow y$, $z_2 \rightarrow \dot{y}$, $z_3 \rightarrow d$.

3.3. Position–Velocity Integrated Controller Design

To achieve a fast response and strong robustness in the servo system, a GNFTSMS based on the reconstructed error is designed, combining the estimated disturbance for feed-forward compensation to realize the design of a position–velocity integrated controller. The position tracking error is defined as:

$$e_1 = \theta^* - \theta \quad (20)$$

$$e_2 = \dot{e}_1 \quad (21)$$

where θ^* represents the reference position of the rotor, and θ denotes the actual rotor position. Substituting (20) into (11), the reconstructed error is obtained, and its first derivative and second derivative are as follows:

$$\dot{\varepsilon} = \frac{1}{2} \left[\frac{1}{z(t) + \underline{\delta}} - \frac{1}{z(t) - \bar{\delta}} \right] \left(\frac{\dot{e}_1}{\mu} - \frac{e_1 \dot{\mu}}{\mu^2} \right) = r \left(\dot{e}_1 - \frac{e_1 \dot{\mu}}{\mu} \right) \quad (22)$$

$$\begin{aligned} \ddot{\varepsilon} &= \dot{r} \left(\dot{e}_1 - \frac{e_1 \dot{\mu}}{\mu} \right) + r \left[\ddot{e}_1 - \frac{(\dot{e}_1 \dot{\mu} + e_1 \ddot{\mu})\mu - e_1 \dot{\mu}^2}{\mu^2} \right] \\ &= \dot{r} \left(\dot{e}_1 - \frac{e_1 \dot{\mu}}{\mu} \right) - r \frac{\dot{e}_1 \dot{\mu} \mu + e_1 \ddot{\mu} \mu - e_1 \dot{\mu}^2}{\mu^2} + r \ddot{e}_1 \\ &= R + r \ddot{e}_1 \end{aligned} \quad (23)$$

where $r = \frac{1}{2\mu} \left[\frac{1}{z(t) + \underline{\delta}} - \frac{1}{z(t) - \bar{\delta}} \right]$, $R = \dot{r} \left(\dot{e}_1 - \frac{e_1 \dot{\mu}}{\mu} \right) - r \frac{\dot{e}_1 \dot{\mu} \mu + e_1 \ddot{\mu} \mu - e_1 \dot{\mu}^2}{\mu^2}$.

The GNFTSMS is designed as:

$$s = \dot{\varepsilon} + c_1 \varepsilon + c_2 e^{-\lambda t} \varepsilon^{(1-2\beta)} \quad (24)$$

$$\dot{\varepsilon} = -c_1 \varepsilon - c_2 e^{-\lambda t} \varepsilon^{(1-2\beta)} \quad (25)$$

where c_1 , c_2 , and λ are positive constants with $0 < \beta < 1$.

From this form of the sliding surface, it can be observed that the time-varying component $e^{-\lambda t}$ will converge to 0 after a certain period. At this point, the sliding surface will transition to a linear form. By selecting an appropriate value of λ , the designed sliding surface will possess the dual advantages of both a non-singular terminal sliding mode surface and a linear sliding mode surface.

For the PMSM servo system, the designed sliding mode surface ensures that when $2\beta c_1 - \lambda > 0$ is satisfied, the system reconstructed error can converge to 0 within a finite time. The convergence time for this type of sliding mode surface is proven in [11] as:

$$T_s \leq \frac{\ln(1 + \frac{e^{2\beta c_1 t} V^\beta(0)}{a_2})}{2\beta c_1 - \lambda} \quad (26)$$

Theorem 1. When $\bar{\delta} = \underline{\delta}$, ε and e_1 exhibit similar convergence behavior. If the reconstructed error ε converges to 0 within a finite time, the original system error e_1 will also converge to 0 within a finite time.

Proof. According to Equations (9) and (20), $\varepsilon \rightarrow 0$ implies:

$$e_1 = \mu(t)S(0) = \mu(t) \left(\frac{\bar{\delta} - \underline{\delta}}{2} \right) \quad (27)$$

Due to $\mu(t) \neq 0$, the condition $\bar{\delta} = \underline{\delta}$ results in

$$e_1 \rightarrow 0 \quad (28)$$

Here, the proof is complete. $\square$

Taking the derivative of the sliding surface gives:

$$\begin{aligned}\dot{s} &= \ddot{\varepsilon} + c_1\dot{\varepsilon} + c_2e^{-\lambda t}\varepsilon^{-2\beta}[-\lambda\varepsilon + (1 - 2\beta)\dot{\varepsilon}] \\ &= R + r\ddot{\varepsilon} + c_1\dot{\varepsilon} + C \\ &= R + r(\ddot{\theta}^* - bi_q - d) + c_1\dot{\varepsilon} + C\end{aligned}\quad (29)$$

where $C = c_2e^{-\lambda t}\varepsilon^{-2\beta}[-\lambda\varepsilon + (1 - 2\beta)\dot{\varepsilon}]$.

The exponential reaching law is adopted as:

$$\dot{s} = -q\text{sign}(s) - ks \quad (30)$$

Combining the sliding surface dynamics with feed-forward compensation, the position–velocity integrated controller is designed as:

$$i_q^* = \frac{1}{b}[\ddot{\theta}^* - z_3 + \frac{1}{r}(R + c_1\dot{\varepsilon} + C + q\text{sign}(s) + ks)] \quad (31)$$

To ensure system stability under this control law, the following Lyapunov function is constructed:

$$V_1 = \frac{1}{2}s^2 \quad (32)$$

Based on the above expression, we obtain:

$$\dot{s} = -ks - q\text{sign}(s) + z_3 - d \quad (33)$$

where z_3 represents the real-time estimate of the disturbance d obtained via the LESO. By introducing the observation error function $z_3 - d$, the effect of the disturbance d is dynamically compensated. From Equation (19), it follows that $z_3 - d \rightarrow 0$, which leads to:

$$\dot{s} = -ks - q\text{sign}(s) \quad (34)$$

$$\dot{V}_1 = s\dot{s} = -ks^2 - q|s| \leq 0 \quad (35)$$

Thus, the stability of the system is ensured within the Lyapunov function framework.

3.4. Current Controller Design

The q-axis and d-axis current controllers both employ PI control. The q-axis PI controller generates the q-axis reference voltage by utilizing the error between the reference current, generated by the position–velocity integrated controller, and the actual feedback current. The d-axis PI controller generates the d-axis reference voltage by utilizing the error between the set reference current $i_d^* = 0$ and the actual feedback current. The voltage is then processed through the inverse Park transformation and SVPWM, producing the inverter control signals, which in turn control the motor and complete the position tracking control of the servo system. The process can be described by the following equations:

$$e_3 = i_q^* - i_q, \quad e_4 = i_d^* - i_d \quad (36)$$

$$u_q = K_{p,q}e_3 + K_{i,q} \int e_3 dt \quad (37)$$

$$u_d = K_{p,d}e_4 + K_{i,d} \int e_4 dt \quad (38)$$

where $K_{p,q}$ and $K_{p,d}$ are the proportional gains of the q-axis and d-axis current controllers, respectively; $K_{i,q}$ and $K_{i,d}$ are the integral gains of the q-axis and d-axis current controllers, respectively.

3.5. Practical Implementation

The implementation of the proposed method follows these steps:

- (1) Design the initial parameters, as detailed in the subsequent section;
- (2) Calculate the position tracking error, obtain the reconstructed error and its derivative, and derive the GNFTSMS;
- (3) Compute the actual control action based on Equation (31);
- (4) Record input and output measurements according to the instructions in Section 3.4;
- (5) Return to (2) for the next sampling.

4. Analysis of Experimental Results

In this study, both simulation and experimental tests were conducted to validate the effectiveness and performance of the proposed method. The simulation focused on step signal tracking and step signal tracking under sudden load disturbances. Six comparative control methods were employed: PID, SMC, GNFTSMC, SMC-LESO, GNFTSMC-LESO, and the proposed method. The control laws for each method are as follows:

$$w^* = Ke_1, \quad e_5 = w^* - w, \quad u_{PI} = K_p e_5 + K_i \int e_5 dt \quad (39)$$

$$u_{SMC} = \frac{1}{b} [\ddot{\theta}^* + c_1 e_2 + q \operatorname{sign}(s) + ks] \quad (40)$$

$$u_{GNFTSMC} = \frac{1}{b} [\ddot{\theta}^* + c_1 e_2 + A + q \operatorname{sign}(s) + ks] \quad (41)$$

$$u_{SMC-LESO} = \frac{1}{b} [\ddot{\theta}^* + c_1 e_2 + q \operatorname{sign}(s) + ks - z_3] \quad (42)$$

$$u_{GNFTSMC-LESO} = \frac{1}{b} [\ddot{\theta}^* + c_1 e_2 + A + q \operatorname{sign}(s) + ks - z_3] \quad (43)$$

where K is the proportional gain for the position controller, and K_p and K_i are the proportional and integral gains for the speed controller, respectively. $A = c_2 e^{-\lambda t} e_1^{-2\beta} [(1 - 2\beta)e_2 - \lambda e_1]$.

The PMSM parameters and the settings for each controller used in the simulation are listed in Tables 1 and 2. All controller parameters were selected through empirical tuning to balance response time and steady-state accuracy, with the tuning process primarily based on trial and error.

Table 1. Simulation parameters of PMSM.

Parameter	Value
Stator inductance	8.5 mH
Stator resistance	2.875 Ω
Viscous friction coefficient	$8 \times 10^{-3} \text{ kg} \cdot \text{m}^2/\text{s}$
Number of pole pairs	4
Flux linkage	0.175 Wb
Moment of inertia	$0.003 \text{ kg} \cdot \text{m}^2$

Table 2. Simulation controller parameters.

Methods	Parameters
PID	$K = 6, K_p = 20, K_i = 65$
SMC	$c_1 = 5, q = 35, k = 50$
GNFTSMC	$c_1 = 5, c_2 = 0.1, \lambda = 2, \beta = 0.3, q = 35, k = 50$
SMC-LESO	$c_1 = 5, q = 5, k = 200, w_0 = 100$
GNFTSMC-LESO	$c_1 = 5, c_2 = 0.1, \lambda = 2, \beta = 0.3, q = 5, k = 200, w_0 = 100$
Proposed	$\mu_0 = 3.5, \mu_\infty = 0.02, \bar{\delta} = \underline{\delta} = 1, l = 5, c_1 = 5, c_2 = 0.1, \lambda = 2, \beta = 0.3, q = 5, k = 200, w_0 = 100$

Simulation 1: Step signal tracking with an amplitude of π rad. Figure 3a,c show the system position tracking and tracking error curves, respectively. Experimental results indicate that traditional PID and SMC control methods exhibit an overshoot and slow response during the dynamic phase, failing to quickly reach a steady state. In contrast, the GNFTSMC method shows an improved response speed but still experiences significant tracking errors in the steady-state phase. On the other hand, the three LESO-based control algorithms demonstrate a notable advantage by slightly accelerating the response speed while maintaining minimal steady-state errors, validating the effectiveness of the real-time disturbance compensation algorithm. Among them, the proposed method has the fastest convergence speed, outperforming the others in both dynamic and steady-state performance, achieving high-precision tracking in a shorter time. Figure 3e shows the system control input curves. To ensure fairness, the control input is limited to $\pm 30A$, and the current controller parameters are consistent across all control methods, specifically: $K_{p,q} = K_{p,d} = 17, K_{i,q} = K_{i,d} = 5750$. The zoomed-in view reveals that PID control results in fewer current oscillations compared to the traditional SMC method, exhibiting better stability, second only to the proposed method. Traditional SMC and GNFTSMC methods, lacking effective suppression, exhibit noticeable current oscillations. The LESO-based control algorithms significantly smooth the control input signal, with SMC-LESO and GNFTSMC-LESO showing better current stability, while the proposed method exhibits the smoothest control current. This highlights its superior ability in suppressing oscillations and improving system stability. The integration of PPC and the observer not only enhances the system dynamic performance but also improves control input smoothness, making the system operation more reliable.

Simulation 2: Step signal tracking with an amplitude of π rad, with a load torque of $0.3 \text{ N} \cdot \text{m}$ applied at 1.5 s. Figure 3b,d show the system position tracking and tracking error curves, respectively. Experimental results show that LESO-based control methods have significant advantages in both position tracking accuracy and convergence speed, with the proposed method being particularly effective in quickly reaching equilibrium and achieving high-precision tracking. Among all tested methods, the SMC and GNFTSMC methods exhibit smaller transient errors under disturbance but suffer from larger steady-state errors due to their reliance on disturbance upper bound compensation. The PID method, while tracking the desired position well and performing reasonably in disturbance suppression, has a slower convergence speed and fails to promptly reach a steady state. In contrast, LESO-based methods estimate disturbances in real time, leading to faster convergence and smaller steady-state errors. Notably, the proposed method, despite having a larger transient error under step-load disturbance, exhibits the best overall performance, though further optimization is still possible. Figure 3f shows the system control input curves. For fairness, the control input limit and controller parameters are consistent with those used in step signal tracking. The zoomed-in view indicates that the disturbance causes a jump in the control current. For the SMC and GNFTSMC methods, which use larger

switching gains to compensate for disturbances, current oscillations are more noticeable before the disturbance and decrease afterward. The PID, SMC-LESO, and GNFTSMC-LESO methods significantly reduce current oscillations, while the proposed method consistently maintains the smoothest control current, demonstrating its superior ability to suppress current fluctuations and enhance system stability.

To further validate the proposed algorithm, a PMSM experimental platform was constructed, illustrated in Figure 4. Two 42JSF630AS-1000 PMSMs were selected for safety and performance considerations. These two motors were connected into a pair of drag motors via a coupling and mounting bracket, with one functioning as the driving motor and the other as the load motor. The control and drive components were assembled in a control box, using the Texas Instruments DSP28379D and DRV8305evm (Dallas, TX, USA) to meet the real-time requirements of complex control algorithms and ensure system stability and safety.

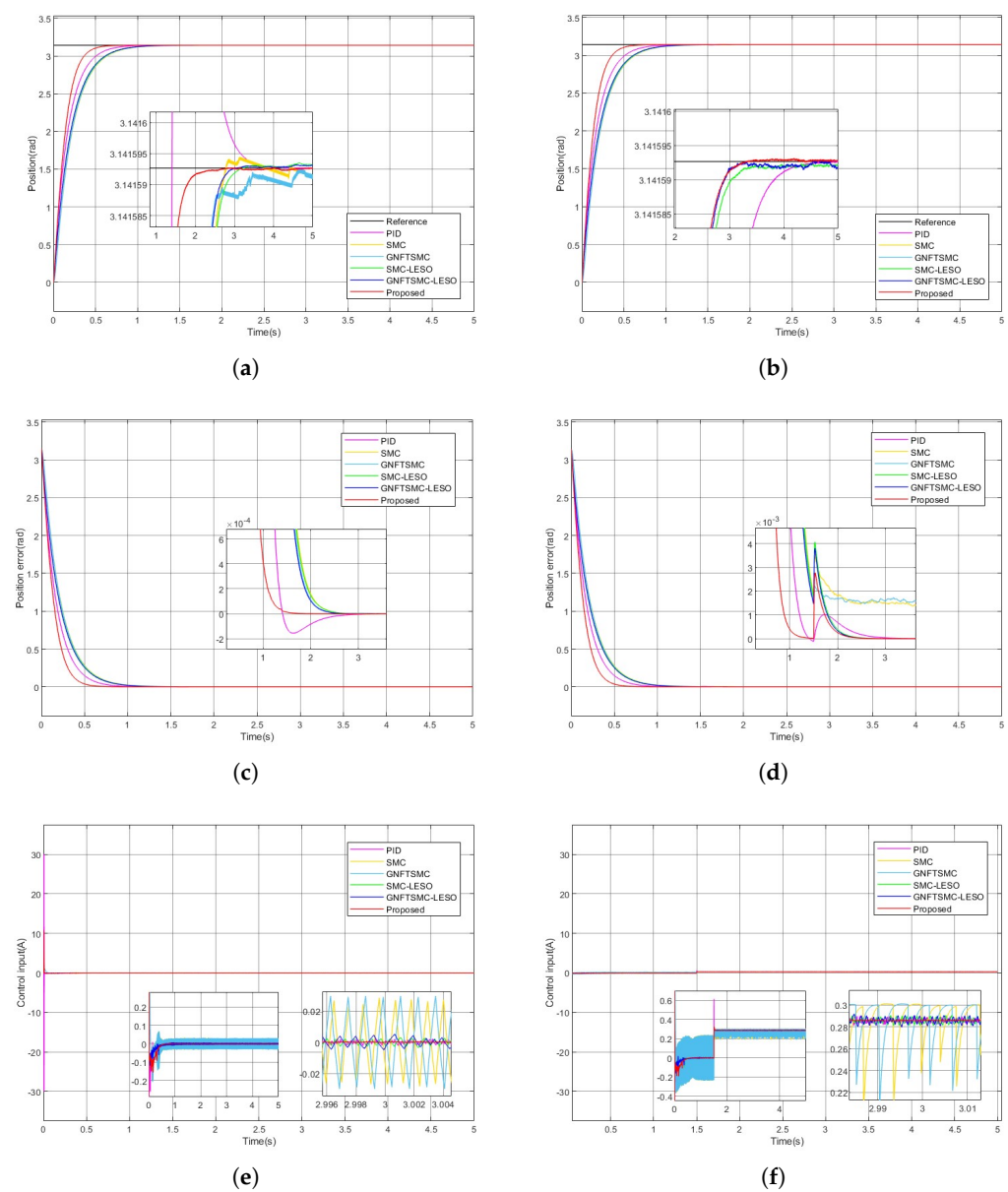


Figure 3. System performance curves of simulations. (a) Position tracking curves. (b) Position tracking curves. (c) Position tracking error curves. (d) Position tracking error curves. (e) Control input curves. (f) Control input curves.

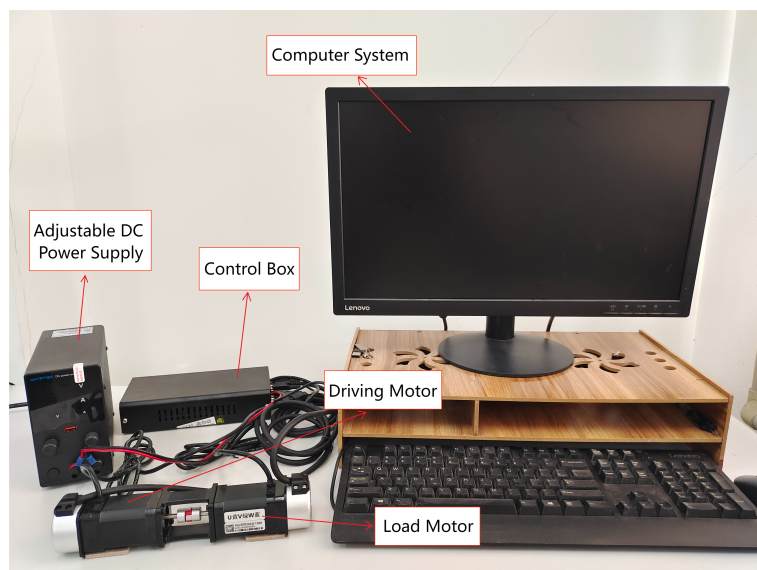


Figure 4. Experiment platform.

The system was connected directly to a computer, allowing for real-time data monitoring and parameter adjustment, as well as enabling the transmission of control commands and reception of feedback data during experiments. This setup facilitates algorithm optimization and performance analysis. The entire system was powered by a 24 V adjustable DC power supply. The relevant parameters of the PMSM used in the experiment are listed in Table 3.

Table 3. Experimental parameters of PMSM.

Parameter	Value
Stator inductance	0.62 mH
Stator resistance	0.89 Ω
Viscous friction coefficient	$3.5 \times 10^{-4} \text{ kg} \cdot \text{m}^2/\text{s}$
Number of pole pairs	4
Flux linkage	0.0173 Wb
Moment of inertia	$2.8 \times 10^{-6} \text{ kg} \cdot \text{m}^2$

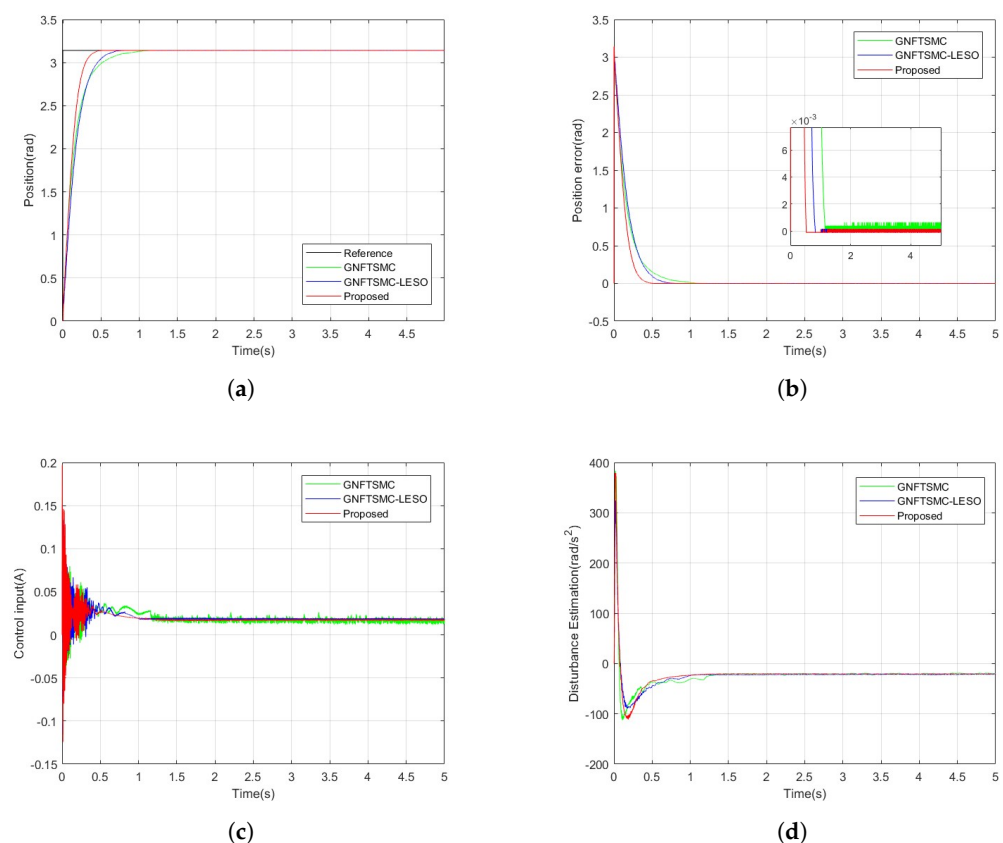
Four experimental scenarios were designed: step signal tracking, sine wave signal tracking, triangular wave signal tracking, and step signal tracking with a sudden load disturbance. In the step signal tracking experiment, the parameters for all tested methods are presented in Table 4. For the proposed method and GNFTSMC-LESO method, the parameter settings were identical, while for the GNFTSMC method, the only difference was $q = 35$. For the remaining experiments, certain parameters were adjusted to ensure fairness in the comparison, keeping all controllers under similar disturbance conditions, as validated in the subsequent analysis. The current controller parameters used in all methods were: $K_{p,q} = K_{p,d} = 3$, $K_{i,q} = K_{i,d} = 0.28$. The control input saturation limit was set to $\pm 3A$. The position–velocity loop was sampled at 10 kHz, the current loop at 20 kHz, and the PWM frequency was fixed at 20 kHz.

The experimental tests employed three methods: GNFTSMC, GNFTSMC-LESO, and the proposed method. Based on Equations (31), (41), and (43), it can be observed that each controller contains a time-dependent function. The time is calculated as follows: when the motor is enabled and the system is in closed-loop operation, the value 1 is integrated to serve as the time input to the controller.

Table 4. Experimental controller parameters.

Methods	Parameters
GNFTSMC	$c_1 = 8, c_2 = 0.1, \lambda = 4, \beta = 0.6, q = 35, k = 50$
GNFTSMC-LESO	$c_1 = 8, c_2 = 0.1, \lambda = 4, \beta = 0.6, q = 2, k = 50, w_0 = 75$
Proposed	$\mu_0 = 4, \mu_\infty = 0.15, \bar{\delta} = \underline{\delta} = 1, l = 4.5, c_1 = 8, c_2 = 0.1, \lambda = 4, \beta = 0.6, q = 2, k = 50, w_0 = 75$

Experiment 1: Step signal tracking with an amplitude of π rad, simulating a scenario where rapid position changes are required for point control in CNC machines. The system position tracking curves are shown in Figure 5a. Overall, all methods track the given position signal effectively, but the proposed method exhibits a faster response time, reaching equilibrium in a shorter time. Figure 5b shows the position tracking error curves, where the inclusion of the observer significantly improves tracking accuracy, reducing both error and chattering. Figure 5c shows the system control input curves, demonstrating that the proposed method exhibits superior performance with smaller fluctuations in control current, resulting in a smoother control effect. Figure 5d shows the disturbance estimation curves. Although the GNFTSMC method does not include an observer for disturbance compensation, the same disturbance estimation analysis was conducted for fairness. The curve indicates that while the observer experiences some error during the initial phase, it quickly stabilizes over time, showing that the observer can accurately estimate disturbances in a short period.

**Figure 5.** System performance curves of Experiment 1. (a) Position tracking curves. (b) Position tracking error curves. (c) Control input curves. (d) Disturbance estimation curves.

Experiment 2: Sine signal tracking with an amplitude of 1 rad, simulating a scenario where smooth and continuous position changes are required for trajectory control in CNC machines. Figure 6a,b show the system position tracking and tracking error curves,

respectively. Overall, all methods track the given position signal effectively. By comparing the detailed position tracking curves with the tracking error curves, it is evident that the GNFTSMC method shows a relatively small tracking error initially. However, as the tracking error of the proposed method gradually approaches zero, the CNFTSMC method exhibits significant error, which persists throughout the entire process. In contrast, both the GNFTSMC-LESO and proposed methods exhibit slightly larger tracking errors at the initial stage and peak points, while the errors remain small and fluctuate within a narrow range during the rest of the process, further confirming the superiority of the observer-based compensation method. Figure 6c shows the system control input curves, with the proposed method exhibiting smoother control input and significantly reduced fluctuations in control current. Figure 6d shows the disturbance estimation curves. Although some disturbances were present in the physical experiment, the disturbance environment was kept as consistent as possible to ensure a fair comparison.

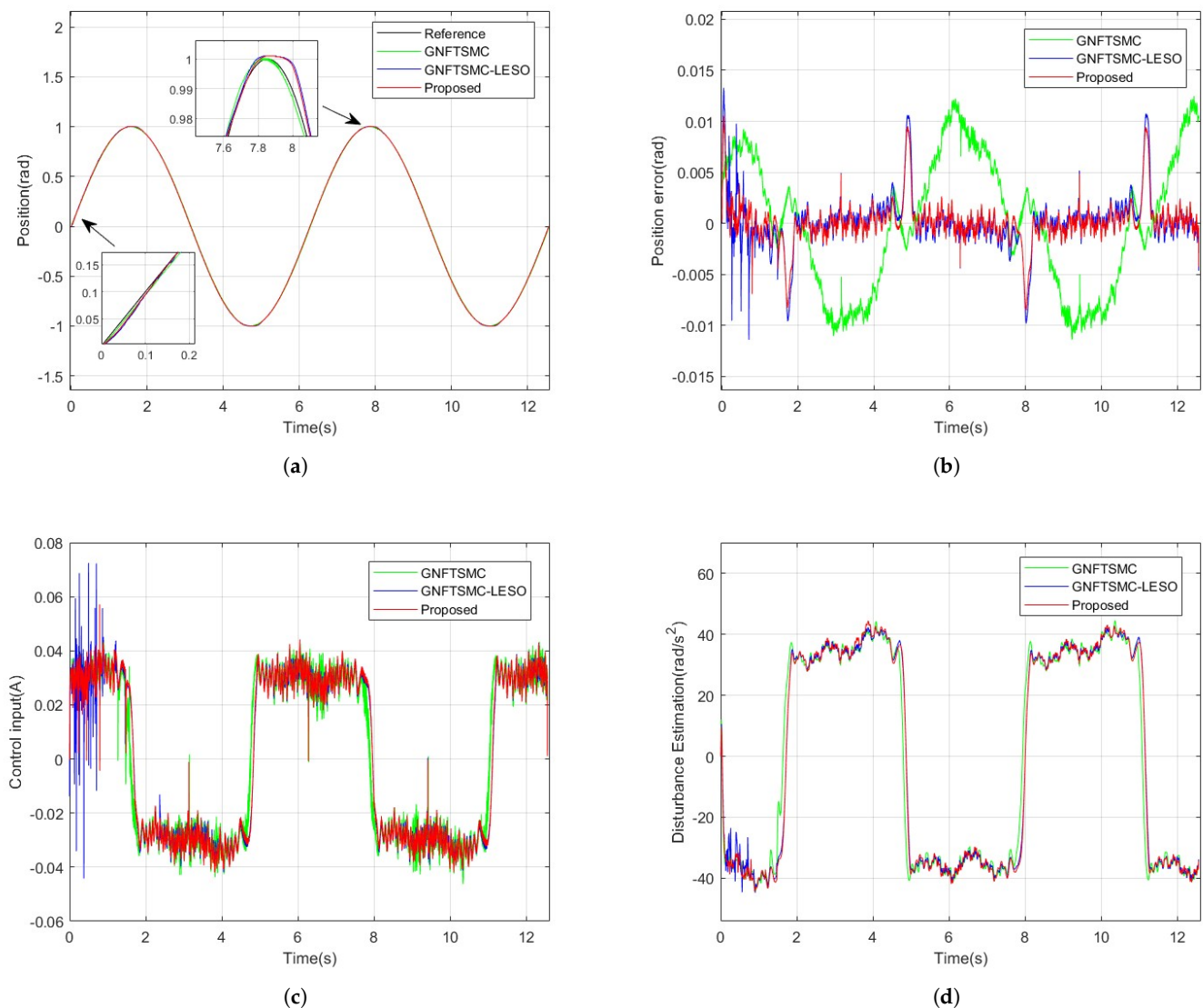


Figure 6. System performance curves of Experiment 2. (a) Position tracking curves. (b) Position tracking error curves. (c) Control input curves. (d) Disturbance estimation curves.

Experiment 3: Triangular wave signal tracking with an amplitude of 1 rad, simulating a scenario where periodic motion or position changes are required for straight-line control operations in CNC machines. Figure 7a,b show the system position tracking and tracking

error curves, respectively. Overall, all methods track the given position signal effectively. By comparing the detailed zoomed-in figures and tracking error curves, it is evident that the proposed method responds more quickly in the initial stage and exhibits smaller tracking errors. Furthermore, this method only shows slightly larger errors at the peak points, with the errors fluctuating close to zero for the rest of the process, reflecting its high tracking accuracy and stability. Figure 7c shows the system control input curves, with the proposed method exhibiting smoother control input in the initial stage and significantly reduced current fluctuations. In subsequent periods, the control input is comparable to that of the other methods. Figure 7d shows the disturbance estimation curves. To ensure experimental fairness, the disturbance environment was kept as consistent as possible. The observer in the proposed method can accurately estimate the disturbance and quickly converge to a stable state.

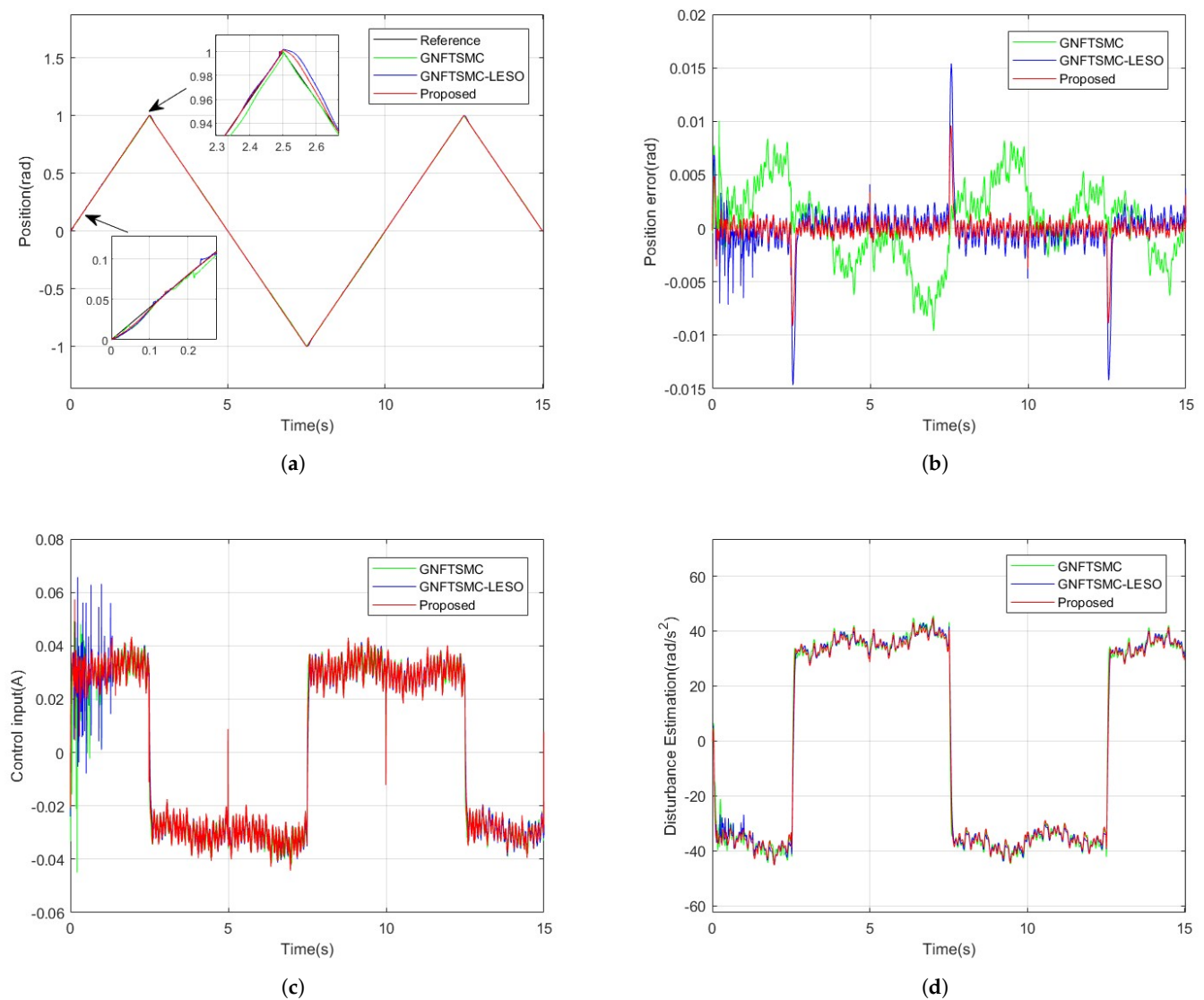


Figure 7. System performance curves of Experiment 3. (a) Position tracking curves. (b) Position tracking error curves. (c) Control input curves. (d) Disturbance estimation curves.

Experiment 4: Step signal tracking with an amplitude of π rad, with a load torque of $0.05 \text{ N} \cdot \text{m}$ applied at 1.36 s. Figure 8a,b show the system position tracking and tracking error curves, respectively. Overall, all methods track the given position signal effectively. Zooming in on the error curves reveals that the GNFTSMC method, based on upper-

bounding disturbance compensation, is least affected by the disturbance. However, this method requires prior knowledge of the disturbance upper bound, which is often hard to obtain accurately in real-world environments. Additionally, it induces significant system chattering, limiting its practical applicability. In contrast, both the GNFTSMC-LESO and proposed methods estimate disturbances in real time using an observer. While they exhibit some tracking errors, these remain within a small range. The proposed method converges faster and is less affected by the disturbance. Figure 8c shows the system control input curves. It is evident that the proposed method produces smoother control inputs and smaller current fluctuations, which improves the system's practical performance. Figure 8d shows the disturbance estimation curves. It is indicated that the observer accurately estimates the disturbance and converges to a stable state in a shorter time, though its convergence speed could be further optimized. Additionally, the maximum position error and settling time under sudden load torque for each method were quantified, as shown in Table 5, further validating our previous analysis.

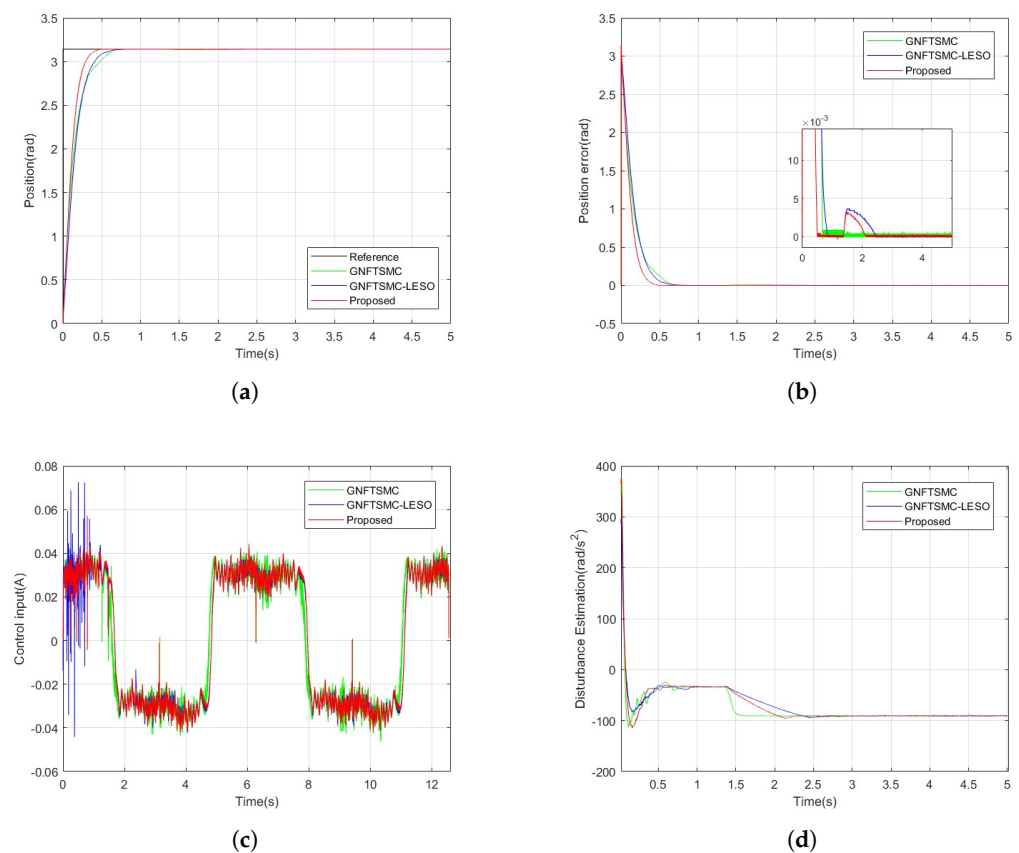


Figure 8. System performance curves of Experiment 4. (a) Position tracking curves. (b) Position tracking error curves. (c) Control input curves. (d) Disturbance estimation curves.

Table 5. Maximum position error and settling time under sudden load torque in Experiment 4.

Index	GNFTSMC	GNFTSMC-LESO	Proposed
Maximum position error (rad)	0.0006	0.0037	0.0032
Settling time (s)	0.0382	1.1002	0.7281

To evaluate the proposed method more thoroughly, tracking errors for different signal types were quantified. For step signals, the mean absolute error (MAE) and root mean square error (RMSE) were adopted for evaluation, while for sine and triangular wave

signals, the maximum error (MAX), MAE, and RMSE were utilized. These error metrics are defined as follows:

$$\text{MAX} = \max(|e_1|, |e_2|, \dots, |e_N|) \quad (44)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |e_i| \quad (45)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2} \quad (46)$$

where e_i is the error at the i -th sampling point.

The quantified tracking error data are shown in Table 6, indicating that the proposed method demonstrates the smallest tracking errors across all experiments, further confirming its superiority in control accuracy, response speed, and disturbance rejection.

Table 6. Quantified tracking error data.

Experiment	Index	GNFTSMC	GNFTSMC-LESO	Proposed
1	MAE (rad)	0.1003	0.0913	0.0606
	RMSE (rad)	0.3908	0.3947	0.3182
2	MAX (rad)	0.0125	0.0133	0.0105
	MAE (rad)	0.0054	0.0015	0.0012
	RMSE (rad)	0.0064	0.0025	0.0021
3	MAX (rad)	0.0100	0.0154	0.0096
	MAE (rad)	0.0030	0.0012	0.0006
	RMSE (rad)	0.0037	0.0023	0.0013
4	MAE (rad)	0.0961	0.0965	0.0733
	RMSE (rad)	0.3916	0.4128	0.3641

Based on the comparison between the experimental and simulation results, we observed certain discrepancies. To explain these differences, we considered possible causes, focusing on both the method itself and the influence of motor parameters. The motor used in the simulation differs from the actual motor in terms of parameters, especially in the values of the moment of inertia and magnetic flux linkage. Specifically, the magnetic flux linkage used in the simulation is 10 times that of the actual motor, and there is also a significant difference in the moment of inertia. These differences are likely the key factors contributing to the differences between the experimental and simulation results.

Among these factors, the difference in the moment of inertia was found to have the most significant impact, as it directly affects the system dynamic response. Therefore, we chose to focus on adjusting only the moment of inertia in the subsequent experiments. While the difference in magnetic flux linkage might also affect the control performance, its impact on the steady-state performance of the system is relatively limited compared to that of the moment of inertia. As a result, we decided not to adjust the magnetic flux linkage in the experiments. By concentrating on variations in the moment of inertia, we can avoid the potential confounding effects caused by varying multiple factors simultaneously.

To this end, a series of experiments were conducted, keeping all motor parameters except the moment of inertia fixed to the actual motor values. Simulations were carried out with moment of inertia values of 0.028×10^{-4} , 0.028×10^{-3} , and 0.028×10^{-2} . The experimental results, as shown in Figure 9, indicate that as the moment of inertia increases, the experimental results gradually approach the simulation expectations. This suggests that the proposed method may be more suitable for motors with larger moments of inertia, or may require a specific ratio of magnetic flux linkage to moment of inertia.

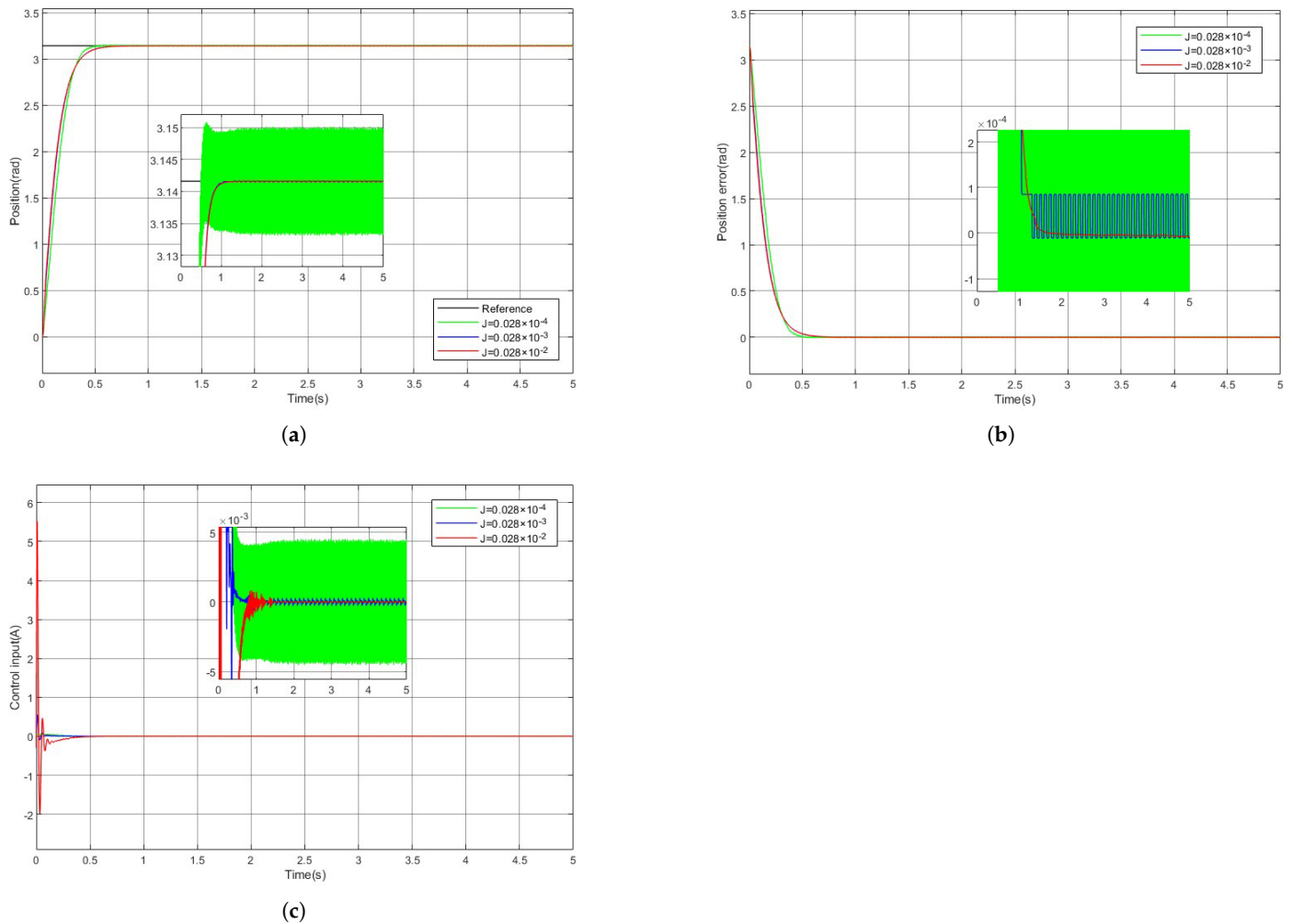


Figure 9. Effect of moment of inertia on experimental results. (a) Position tracking curves. (b) Position tracking error curves. (c) Control input curves.

We also considered that the use of the sign function in the controller could contribute to the observed deviation between the actual control performance and the simulation results. The nonlinear characteristics introduced by the sign function, particularly in the actual system, may negatively affect the control accuracy. To mitigate the impact of this nonlinearity, we plan to optimize the controller in the future by adopting saturation functions or other smooth functions, which will improve the system robustness and precision. Additionally, we will continue to investigate the impact of the moment of inertia on the performance of the control method and validate its adaptability under different motor parameters.

5. Conclusions

This paper proposes a LESO-based PPGNFTSMC method to address challenges in maintaining position tracking accuracy and system stability, which are often impacted by uncertainties in practical manufacturing processes. A series of experiments were conducted, under different conditions, including step, sine wave, and triangular wave signals, along with detailed error quantification analysis for each condition. The results demonstrate that the proposed method outperforms the comparison methods in PMSM position tracking control. Specifically, it shows stronger robustness, a faster response, and enhanced disturbance rejection. Thus, it is especially suitable for CNC machine tools requiring high-precision position tracking, providing an effective solution to the challenges faced

by high-end CNC machine tools in complex environments. Future research will focus on further optimizing the proposed method and exploring the adaptive parameter adjustment mechanism. This mechanism will enable real-time dynamic adjustments based on varying operational conditions and environmental changes, thereby enhancing its adaptability and robustness for a broader range of application scenarios.

Author Contributions: Conceptualization, Y.Y., Y.Z., Y.X. and P.X.; methodology, Y.Y.; validation, Y.Y.; resources, Y.Z. and C.W.; writing—original draft preparation, Y.Y.; writing—review and editing, Y.Z.; funding acquisition, Y.Z. and C.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported in part by the Ministry of Industry and Information Technology Project (TC220H05X-04), the National Natural Science Foundation of China (62403108, U20A20197 and 61973063), the Liaoning Provincial Natural Science Foundation Joint Fund (2023-MSBA-075), the Scientific Research Foundation of Liaoning Provincial Education Department (LJKQR20222509), and the Fundamental Research Funds for the Central Universities (N2426005).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

PMSM	permanent magnet synchronous motor
PPGNFTSMC	prescribed performance global non-singular fast terminal sliding mode control
LESO	linear extended state observer
CNC	computer numerical control
GNFTSMC	global non-singular fast terminal sliding mode control
GNFTSMC-LESO	GNFTSMC with LESO
SMC	sliding mode control
NFTSMC	non-singular fast terminal sliding mode control
PPC	prescribed performance control
PI	proportional–integral
GNFTSMS	global non-singular fast terminal sliding mode surface
PPF	prescribed performance function
PWM	pulse width modulation
MAE	mean absolute error
RMSE	root mean square error
MAX	maximum error

References

1. Zhu, K.; Zhang, Y. A Cyber-Physical Production System Framework of Smart CNC Machining Monitoring System. *IEEE ASME Trans. Mechatron.* **2018**, *23*, 2579–2586.
2. Sun, X.; Chen, L.; Yang, Z. Overview of Bearingless Permanent-Magnet Synchronous Motors. *IEEE Trans. Ind. Electron.* **2013**, *60*, 5528–5538.
3. Sira-Ramirez, H.; Linares-Flores, J.; Garcia-Rodriguez, C.; Contreras-Ordaz, M. A. On the Control of the Permanent Magnet Synchronous Motor: An Active Disturbance Rejection Control Approach. *IEEE Trans. Control Syst. Technol.* **2014**, *22*, 2056–2063.
4. Huang, Y.; Zou, J.; Wang, M.; Xu, Y.; Zou, J. Active disturbances rejection controller for position servo control of PMSM. In Proceedings of the 22nd International Conference on Electrical Machines and Systems (ICEMS), Harbin, China, 11–14 August 2019; pp. 1–4.

5. Linares-Flores, J.; Garcia-Rodriguez, C.; Sira-Ramirez, H.; Ramirez-Cardenas, O. D. Robust Backstepping Tracking Controller for Low-Speed PMSM Positioning System: Design, Analysis, and Implementation. *IEEE Trans. Ind. Inform.* **2015**, *11*, 1130–1141.
6. Niu, S.; Luo, Y.; Fu, W.; Zhang, X. Robust Model Predictive Control for a Three-Phase PMSM Motor with Improved Control Precision. *IEEE Trans. Ind. Electron.* **2021**, *68*, 838–849.
7. Ying-Shieh, K.; Chung-Chun, H.; Ming-Hung, T. FPGA Realization of an Adaptive Fuzzy Controller for PMLSM Drive. *IEEE Trans. Ind. Electron.* **2009**, *56*, 2923–2932.
8. Gu, J.; You, S.; Kim, W.; Moon, J. Fuzzy Event-Triggered Super Twisting Sliding Mode Control for Position Tracking of Permanent Magnet Synchronous Motors Under Unknown Disturbances. *IEEE Trans. Ind. Inform.* **2023**, *19*, 9843–9854.
9. Shyu, K.-K.; Lai, C.-K.; Tsai, Y.-W.; Yang, D.-I. A Newly Robust Controller Design for the Position Control of Permanent-Magnet Synchronous Motor. *IEEE Trans. Ind. Electron.* **2002**, *49*, 558–565.
10. Li, S.; Zhou, M.; Yu, X. Design and Implementation of Terminal Sliding Mode Control Method for PMSM Speed Regulation System. *IEEE Trans. Ind. Inform.* **2013**, *9*, 1879–1891.
11. Fu, D.; Zhao, X.; Zhu, J. A Novel Robust Super-Twisting Nonsingular Terminal Sliding Mode Controller for Permanent Magnet Linear Synchronous Motors. *IEEE Trans. Power Electron.* **2022**, *37*, 2936–2945.
12. Ding, B.; Xu, D.; Jiang, B.; Shi, P.; Yang, W. Disturbance-Observer-Based Terminal Sliding Mode Control for Linear Traction System With Prescribed Performance. *IEEE Trans. Transp. Electrification* **2021**, *7*, 649–658.
13. Dong, H.; Yang, X.; Gao, H.; Yu, X. Practical Terminal Sliding-Mode Control and Its Applications in Servo Systems. *IEEE Trans. Ind. Electron.* **2023**, *70*, 752–761.
14. Wang, Y.; Gu, L.; Xu, Y.; Cao, X. Practical Tracking Control of Robot Manipulators With Continuous Fractional-Order Nonsingular Terminal Sliding Mode. *IEEE Trans. Ind. Electron.* **2016**, *63*, 6194–6204.
15. Yao, X.; Park, J. H.; Dong, H.; Guo, L.; Lin, X. Robust Adaptive Nonsingular Terminal Sliding Mode Control for Automatic Train Operation. *IEEE Trans. Syst. Man Cybern. Syst.* **2019**, *49*, 2406–2415.
16. Solis, C. U.; Clempner, J. B.; Poznyak, A. S. Fast Terminal Sliding-Mode Control With an Integral Filter Applied to a Van Der Pol Oscillator. *IEEE Trans. Ind. Electron.* **2017**, *64*, 5622–5628.
17. Wang, H.; Shi, L.; Man, Z.; Zheng, J.; Li, S.; Yu, M.; Jiang, C.; Kong, H.; Cao, Z. Continuous Fast Nonsingular Terminal Sliding Mode Control of Automotive Electronic Throttle Systems Using Finite-Time Exact Observer. *IEEE Trans. Ind. Electron.* **2018**, *65*, 7160–7172.
18. Xu, D.; Ding, B.; Jiang, B.; Yang, W.; Shi, P. Nonsingular Fast Terminal Sliding Mode Control for Permanent Magnet Linear Synchronous Motor via High-Order Super-Twisting Observer. *IEEE ASME Trans. Mechatron.* **2022**, *27*, 1651–1659.
19. Van, M.; Mavrovouniotis, M.; Ge, S.S. An Adaptive Backstepping Nonsingular Fast Terminal Sliding Mode Control for Robust Fault Tolerant Control of Robot Manipulators. *IEEE Trans. Syst. Man Cybern. Syst.* **2019**, *49*, 1448–1458.
20. Liu, Y.-C.; Laghrouche, S.; Depernet, D.; N'Diaye, A.; Djerdjir, A.; Cirrincione, M. Disturbance-observer-based speed control for SPMSM drives using modified super-twisting algorithm and extended state observer. *Asian J. Control* **2024**, *26*, 1089–1102.
21. Dang, C.; Dou, M.; Yan, S.; Dang, M.; Wang, Y.; Zhao, D.; Hua, Z. An Improved Adaptive Sliding Mode Speed Control of PMSM Drives With an Extended State Observer. *IEEE Trans. Energy Convers.* **2024**, *39*, 2349–2361.
22. Bechlioulis, C.P.; Rovithakis, G.A. Robust Adaptive Control of Feedback Linearizable MIMO Nonlinear Systems with Prescribed Performance. *IEEE Trans. Autom. Control* **2008**, *53*, 2090–2099.
23. Na, J.; Chen, Q.; Ren, X.; Guo, Y. Adaptive Prescribed Performance Motion Control of Servo Mechanisms with Friction Compensation. *IEEE Trans. Ind. Electron.* **2014**, *61*, 486–494.
24. Liang, J.; Chen, Y.; Lai, N.; He, B.; Miao, Z.; Wang, Y. Low-Complexity Prescribed Performance Control for Unmanned Aerial Manipulator Robot System Under Model Uncertainty and Unknown Disturbances. *IEEE Trans. Ind. Inform.* **2022**, *18*, 4632–4641.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.